Started on	Friday, 7 August 2026, 10:36 PM
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Question 1 | Correct Marked out of 1.00

Dinesh is developing a geometry calculator program for a Math class. He wants to write a program that allows users to calculate the area of different geometric shapes such as circles, rectangles, and triangles.

Users can select the type of shape they want to calculate the area for, and the program will perform the corresponding calculation using a 'switch-case' statement.

Formulas Used:

Area of Circle = 3.14 * radius * radius

Area of Rectangle = length * width

Area of Triangle = 0.5 * base * height

Input Format :

The first line of input consists of an integer, representing the geometric shape the user wants to calculate the area for (1 for a circle, 2 for a rectangle, and 3 for a triangle).

1. If the choice is 1, the second line of input consists of a double value **R**, representing the radius of the circle.
2. If the choice is 2, the second line and third lines of input consist of double values **L** and **B**, representing the length and width of the rectangle.
3. If the choice is 3, the second and third lines of input consist of double values **P** and **Q** representing the base and height of the triangle.

Output Format :

If the valid choice is entered, the output prints "**X sq.units**" where X represents the calculated area of the specific shape, rounded off to two decimal places.

Otherwise, the output prints "**Invalid choice**"

For example:

Input	Result
1 5.1	81.67 sq.units
2 4.3 6.1	26.23 sq.units

Answer: (penalty regime: 0 %)

```
1
2
3
4
5
6
7
8
9
10 choice=int(input())
11 if(choice==1):
12     r=float(input())
13     area=3.14*r*r
14     print("{:.2f} sq.units".format(area))
15 elif(choice==2):
16     l=float(input())
17     b=float(input())
18     area=l*b
19     print("{:.2f} sq.units".format(area))
20 elif(choice==3):
21     p=float(input())
22     q=float(input())
23     area=0.5*p*q
```

```
    print("{:.2f} sq.units".format(area))
else:
    print("Invalid choice")

```

	Input	Expected	Got	
✔	1 5.1	81.67 sq.units	81.67 sq.units	✔
✔	2 4.3 6.1	26.23 sq.units	26.23 sq.units	✔
✔	3 3.7 5.6	10.36 sq.units	10.36 sq.units	✔
✔	4	Invalid choice	Invalid choice	✔
✔	1 9.5	283.38 sq.units	283.38 sq.units	✔

Passed all tests! ✔

Question 2 | Correct Marked out of 1.00

Arun is working on a project to automate the process of determining whether a student has passed or failed based on their subject marks.

He aims to create a simple program that takes positive integers as marks for five subjects from the user. If the average of the marks is greater than or equal to 50, the student has passed the exam. Else, the student has failed.

Help Arun to implement the project.

Input Format :
The input consists of five space-separated integers, representing the marks in five subjects.

Output Format :
The first line of output prints "Average score: " followed by an integer representing the average score.

The second line prints one of the following:

1. If the condition is satisfied, print "The student has passed".
2. Otherwise, the output prints "The student has failed".

For example:

Input	Result
50 60 70 80 90	Average score: 70 The student has passed
39 25 30 45 67	Average score: 41 The student has failed

Answer: (penalty regime: 0 %)

```
n1=int(input())
n2=int(input())
n3=int(input())
n4=int(input())
n5=int(input())

avg=(n1+n2+n3+n4+n5)//5

print("Average score:",avg)

if(avg>=50):
    print('The student has passed')
else:
    print("The student has failed")
```

	Input	Expected	Got	
✔	50 60 70 80 90	Average score: 70 The student has passed	Average score: 70 The student has passed	✔
✔	39 25 30 45 67	Average score: 41 The student has failed	Average score: 41 The student has failed	✔

Passed all tests! ✔

Question 3 | Correct Marked out of 1.00

Bruce is working on a task that involves manipulating integers. He needs to rotate the digits of an integer to the right by one position.

Write a program to help Bruce accomplish this task.

Input Format :

The input consists of a single integer **N.**

Output Format :

The output prints the given integer with its digits rotated to the right by one position.

For example:

Input	Result
647	764

Answer: (penalty regime: 0 %)

```
n=input()

if len(n)==1:
    print(n)
else:
    print(n[-1]+n[:-1])
```

	Input	Expected	Got	
✔	647	764	764	✔
✔	78436	67843	67843	✔
✔	82644	48264	48264	✔
✔	62734	46273	46273	✔
✔	2836	6283	6283	✔

Passed all tests! ✔

Question 4 | Correct Marked out of 1.00

Alice, an insurance agent, needs a program to calculate the insurance premium for her clients based on their age and health condition.

The premium amount is determined by the following rules:

1. If the client's age is between 18 and 30 years:
2. 'excellent' health condition: 500.0
3. 'good' health condition: 700.0
4. Any other health condition: 900.0
5. If the client's age is between 31 and 50 years:
6. 'excellent' health condition: 700.0
7. 'good' health condition: 900.0
8. Any other health condition: 1100.0

Write a program that takes the age and health condition of a client as input and outputs the corresponding insurance premium.

Input Format :

The first line of input contains an integer representing the age of the client.

The second line contains a string representing the health condition of the client.

Output Format :

The output prints a double value, representing the insurance premium.

For example:

Input	Result
25 excellent	500.0
31 good	900.0
41 poor	1100.0

Answer: (penalty regime: 0 %)

```
age=int(input())
health=input().strip().lower()

if(age<18):
    print(0.0)

elif(18<=age<=30):
    if(health=="excellent"):
        print(500.0)
    elif(health=="good"):
        print(700.0)
    else:
        print(900.0)


elif(31<=age<=50):
    if(health=="excellent"):
```

```
        print(700.0)
    elif(health=="good"):
        print(900.0)
    else:
        print(1100.0)
```

	Input	Expected	Got	
✔	25 excellent	500.0	500.0	✔
✔	31 good	900.0	900.0	✔
✔	41 poor	1100.0	1100.0	✔
✔	50 good	900.0	900.0	✔

Passed all tests! ✔

Question 5 | Correct Marked out of 1.00

Akash is tasked with developing a program that calculates and categorizes blood pressure based on the given systolic and diastolic readings.

The program should use the following classifications:

1. Low Blood Pressure: Systolic < 90 mm Hg or Diastolic < 60 mm Hg
2. Normal Blood Pressure: Systolic ≤ 120 mm Hg and Diastolic ≤ 80 mm Hg
3. Prehypertension: Systolic ≤ 140 mm Hg and Diastolic ≤ 90 mm Hg
4. Stage 1 Hypertension: Systolic ≤ 160 mm Hg and Diastolic ≤ 100 mm Hg
5. Stage 2 Hypertension: Otherwise

Write a program to assist Akash in computing and classifying blood pressure levels based on input readings.

Input Format :

The input consists of two space-separated integers, representing the systolic blood pressure value **S** and diastolic blood pressure value **D,** respectively.

Output Format :

The output displays "Blood Pressure Category: " followed by the blood pressure category based on the provided input.

Refer to the sample output for the exact text and format.

For example:

Input	Result
50 85	Blood Pressure Category: Low Blood Pressure
112 70	Blood Pressure Category: Normal Blood Pressure

Answer: (penalty regime: 0 %)

```
s=int(input())
d=int(input())

if(s<90 or d<60):
    print("Blood Pressure Category: Low Blood Pressure")
elif(90<s<120 and 60<=d<80):
    print("Blood Pressure Category: Normal Blood Pressure")
elif(120<=s<140 or 80<=d<90):
    print("Blood Pressure Category: Prehypertension")
elif(140<=s<160 or 90<=d<100):
    print("Blood Pressure Category: Stage 1 Hypertension")
else:
    print("Blood Pressure Category: Stage 2 Hypertension")
```

	Input	Expected	Got	
✔	50 85	Blood Pressure Category: Low Blood Pressure	Blood Pressure Category: Low Blood Pressure	✔
✔	112 70	Blood Pressure Category: Normal Blood Pressure	Blood Pressure Category: Normal Blood Pressure	✔
✔	135 86	Blood Pressure Category: Prehypertension	Blood Pressure Category: Prehypertension	✔
✔	145 98	Blood Pressure Category: Stage 1 Hypertension	Blood Pressure Category: Stage 1 Hypertension	✔
✔	170 110	Blood Pressure Category: Stage 2 Hypertension	Blood Pressure Category: Stage 2 Hypertension	✔

Passed all tests! ✔